Corporate Risk Management

Homework #3 Due September 28th, 2026

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Code found at https://github.com/jasonfiammetta/Corporate-Risk/blob/main/hw/hw3.ipynb

At time t we are short M puts and long $M\delta(t, S_t)$ shares of S.
Over period $[t, t+\Delta]$, we hold shares constant, $\lambda = \delta(t, S_t)$.
The values of our portfolio at t and $t+\Delta$ are:

$$V_t = M(\lambda S_t - P^{BS}(t, S_t)) \; ; \; V_{t+\Delta} = M(\lambda S_{t+\Delta} - P^{BS}(t+\Delta, S_{t+\Delta}))$$

1. With $z_t = \ln(S_t)$, identify the full, linearized, and second order loss operators over $[t, t+\Delta]$ as a function of the log return $x = X_{t+\Delta}$

Loss $L = V_t - V_{t+\Delta} = M(\lambda S_t - \lambda S_{t+\Delta} + P^{BS}(t+\Delta, S_{t+\Delta}) - P^{BS}(t, S_t))$

$$z_t = \ln(S_t) \; ; \; x = X_{t+\Delta} = z_{t+\Delta} - z_t = \ln\left(\frac{S_{t+\Delta}}{S_t}\right)$$

$$S_{t+\Delta} = S_t e^x \; ;$$

Full Loss Operator: plug in $S_t e^x$ for $S_{t+\Delta}$,

$$L_{Full}(x) = M(\lambda(S_t - S_t e^x) + P^{BS}(t+\Delta, S_t e^x) - P^{BS}(t, S_t))$$
$$= M(\delta(t, S_t) S_t^{1-e^x} + P^{BS}(t+\Delta, S_t e^x) - P^{BS}(t, S_t))$$

Linearized Loss Operator:

First order Taylor Expansion gives $P^{BS}(t+\Delta, z+x) \approx P^{BS}(t, z) + \frac{\partial P}{\partial t}\Delta + \frac{\partial P}{\partial z}x$

$$\frac{\partial P}{\partial t} = \Theta, \; \frac{\partial P}{\partial z} = S\delta$$

$$P^{BS}(t+\Delta, z+x) - P^{BS}(t, z) \approx \Theta\Delta + S\delta x$$

$$S_{t+\Delta} - S_t = S(e^x - 1) \approx Sx \Rightarrow L_{linear} = M(\delta S x - S\delta x + \Theta\Delta)$$

$$L_{linear}(x) = M\Theta\Delta$$

Second-Order Loss Operator

$$\frac{\partial^2 P}{\partial x^2} = \frac{\partial}{\partial x}\frac{\partial P}{\partial x} = \frac{\partial}{\partial x}\delta S e^x \; ; \text{ at } x=0, \; \frac{\partial}{\partial x}\delta S$$
$$= \gamma S^2 e^{2x} + \delta S e^x \; ; \text{ at } x=0, \; \frac{\partial^2 P}{\partial x^2} = \gamma S^2 + \delta S$$

$$L_{SOL} = M\gamma S^2 = M(\Theta\Delta + \frac{1}{2}\gamma S_t^2 x^2)$$

2. See code for simulations and histograms.

How well do the approximations work?

The linear approximation does not capture the distribution of portfolio losses because delta hedging eliminates the first-order dependence on the stock return, leaving only the deterministic theta component. The second-order approximation does capture the gamma effect and produces a strongly right-skewed loss distribution. In the simulation, the second-order and full loss distributions are very similar, indicating that the second-order Taylor approximation captures most of the portfolio's nonlinear behavior over the ten-day horizon. Differences become more noticeable in the extreme tail, where higher-order terms become important.